

Types of Kernels in Support Vector Machines

IEEE SSCS AUSC — AI Sub-Team · Task 10 / Task 4 — Research Report

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Topic: Kernel methods for SVM classification

Abstract. Support Vector Machines (SVMs) classify data by finding a maximum-margin separator. When classes are not linearly separable in the input space, the kernel trick implicitly maps samples into a higher-dimensional feature space where a linear separator may exist. This report reviews the main SVM kernels — linear, polynomial, Radial Basis Function (RBF / Gaussian), and sigmoid — including their mathematical definitions, hyperparameters, strengths, limitations, and practical selection guidelines.

1. Introduction

A hard-margin SVM solves for a hyperplane $(w^{\top} x + b = 0)$ that separates two classes with the largest possible margin. Soft-margin SVM introduces slack variables and a penalty parameter (C) to allow misclassifications. The dual formulation depends only on inner products $(x_i^{\top} x_j)$. Replacing that inner product with a kernel function $(k(x_i, x_j) = \langle \phi(x_i), \phi(x_j) \rangle)$ yields a non-linear classifier without computing $(\phi(x))$ explicitly — the kernel trick.

For (k) to be a valid Mercer kernel it must be positive semi-definite (Gram matrix of training points has non-negative eigenvalues). Valid kernels can also be combined (e.g. sums and products of kernels remain kernels under mild conditions).

2. The Linear Kernel

$$k(x, x') = x^{\top} x'$$

This is ordinary soft-margin SVM in the original feature space. Hyperparameter focus is mainly (C) (and optionally class weights). Linear SVMs are fast, memory-light, and often strong for high-dimensional sparse data (text, bag-of-words) where the dimensionality itself provides separability.

When to use: baseline model; large (n) or (p) ; interpretability of (w) matters; preliminary check before non-linear kernels.

3. The Polynomial Kernel

$$k(x, x') = (\gamma x^{\top} x' + r)^d$$

Parameters: degree (d) , scale (γ) , and coefficient (r) (often called `coef0`). Degree 2 or 3 captures pairwise / low-order interactions. Higher degrees become expensive and prone to overfitting. Numerical values of $(\gamma x^{\top} x' + r)^d$ can explode for large $(\|x\|)$, so feature scaling is important.

When to use: you expect polynomial interactions of known low order; image patches or engineered features with multiplicative structure. Prefer RBF if you need a flexible non-linear boundary without committing to a degree.

4. The RBF (Gaussian) Kernel

$$k(x, x') = \exp(-\gamma \|x - x'\|^2)$$

Equivalently $\exp(-\|x-x'\|^2 / (2\sigma^2))$ with $\gamma = 1/(2\sigma^2)$. RBF corresponds to an infinite-dimensional feature map. Locality is controlled by γ : large $\gamma \rightarrow$ narrow bumps around support vectors (risk of overfitting); small $\gamma \rightarrow$ smoother, more global influence (risk of underfitting). Together with C , (γ, C) is the usual grid / Optuna search for RBF-SVM.

RBF is the default non-linear choice in practice: it has fewer discrete choices than polynomial degree, is translation-invariant in feature space geometry, and often yields strong accuracy after scaling features to comparable ranges.

When to use: general-purpose non-linear classification on medium-sized tabular or sensor datasets; when linear SVM underfits and you can afford $O(n_{\text{SV}}^2)$ prediction cost.

5. The Sigmoid Kernel

$$k(x, x') = \tanh(\gamma x^\top x' + r)$$

Historically motivated by neural-network activation analogies (two-layer perceptron). It is not always positive semi-definite for arbitrary (γ, r) ; invalid regions can yield non-convex dual problems or unstable solvers. In modern pipelines, sigmoid is used less often than RBF or polynomial, but it remains available in libraries such as scikit-learn for experimentation.

When to use: rarely as a first choice; consider only if domain knowledge suggests tanh-like similarity and parameters are validated carefully.

6. Other Related Kernels (Brief)

- Laplacian: $\exp(-\gamma \|x-x'\|_1)$ — similar spirit to RBF with L1 distance; sometimes more robust to outliers in coordinate noise.
- Chi-square / histogram intersection: common in computer vision for bag-of-visual-words or color histograms.
- String / graph kernels: define similarity on structured objects (DNA, parse trees) without vectorizing by hand.

Custom kernels are admissible if Mercer conditions hold; otherwise solvers may fail or generalization theory no longer applies directly.

7. Hyperparameters and Practical Tuning

Kernel	Key parameters	Notes
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Linear	$\lambda(C)$	Scale features; try <code>class_weight</code> if imbalanced
Polynomial	$\lambda(C, d, \gamma, r)$	Keep $\lambda(d)$ small (2–3); scale inputs
RBF	$\lambda(C, \gamma)$	Log-spaced search; <code>gamma='scale'</code> is a solid default
Sigmoid	$\lambda(C, \gamma, r)$	Validate PSD; prefer RBF unless justified

$\lambda(C)$ trades margin width vs training error: large $\lambda(C) \rightarrow$ harder margin / less regularization. Always standardize or normalize continuous features before non-linear kernels. For multi-class problems, libraries use one-vs-rest or one-vs-one reductions on top of the same kernels.

8. Comparative Discussion

Aspect	Linear	Polynomial	RBF	Sigmoid
Decision boundary	Hyperplane	Polynomial surface	Highly flexible / local	Neural-net-like
Typical speed	Fastest	Medium	Medium–slow	Medium
Overfit risk	Low–medium	High if (d) large	High if (γ) large	Parameter-sensitive
Default recommendation	Strong baseline	Structured interactions	Best general non-linear	Niche

Empirically, a robust workflow is: (1) scale features, (2) fit linear SVM as baseline, (3) fit RBF SVM with nested cross-validation or Optuna over (C, γ) , (4) adopt polynomial only if it matches a known interaction structure. Kernel choice is part of model selection — report validation metrics, not only training accuracy.

9. Connection to This Course Task

In Task 10 we trained both a linear SVM and an RBF SVM on the Red Wine Quality binary classification problem. Linear SVM learns a weighted combination of physicochemical features (alcohol, sulphates, acidity, ...). RBF SVM can bend the boundary around local clusters of “good” vs “bad” wines. Comparing their F1-scores illustrates the bias–variance effect of kernel capacity: RBF often raises training fit and may improve test F1 when the true concept is non-linear, provided (γ) and (C) are not extreme.

10. Conclusion

Kernels are the mechanism that lets SVMs go beyond linear separators while keeping optimization in the dual with inner products. The linear kernel is the right starting point; RBF is the standard non-linear workhorse; polynomial encodes explicit degree- (d) interactions; sigmoid is historically interesting but usually dominated by RBF. Understanding the formula, hyperparameters, and failure modes of each kernel is essential for principled classification with SVMs.

References

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